

MATH230: Tutorial Four [Solutions]
Natural Deductions in First Order Logic

Key ideas

- Read and interpret formulae of first-order logic.
- Write formulae of first-order logic.
- Natural deductions with $\forall \exists$ rules.

Relevant Topic: First-Order Logic.

Relevant reading: Logic and Proof Sections 7,8, and 9.

Hand in exercises: 1a, 1b, 1i, 1j, 1n

Due Friday @ 5pm to the submission box on Learn.

Discussion Questions

1. $\forall x \neg Fx \dashv\vdash \neg \exists x Fx$

Tutorial Exercises

1. Prove the following in the predicate calculus. You will need only the minimal logic rules of inference along with the introduction and elimination rules for the quantifiers.

(a) $\forall x(Fx \rightarrow Gx) \vdash \forall xFx \rightarrow \forall xGx$

$$\frac{\frac{\forall x Fx \rightarrow Gx}{Ft \rightarrow Gt} \forall E \quad \frac{\overline{\forall x Fx}^1}{Ft} \forall E}{\frac{Gt}{\forall x Gx} \forall I} \text{MP} \rightarrow I, 1$$

(b) $\forall x((Fx \vee Gx) \rightarrow Hx), \quad \forall x\neg Hx \vdash \forall x\neg Fx$

$$\frac{\frac{\forall x (Fx \vee Gx \rightarrow Hx)}{Ft \vee Gt \rightarrow Ht} \forall E \quad \frac{\overline{Ft}^1}{Ft \vee Gt} \vee I_L}{\frac{Ht}{\neg Ht} \text{MP}} \frac{\forall x \neg Hx}{\neg Ht} \forall E \rightarrow I, 1$$

$$\frac{\perp}{\neg Ft} \rightarrow I, 1$$

$$\frac{\neg Ft}{\forall x \neg Fx} \forall I$$

(c) $\forall x(Fx \wedge Gx) \vdash \forall xFx \wedge \forall xGx$

$$\frac{\frac{\forall x (Fx \wedge Gx)}{Ft \wedge Gt} \forall E}{\frac{Ft}{\forall x Fx} \forall I} \wedge E_L \quad \frac{\frac{\forall x (Fx \wedge Gx)}{Ft \wedge Gt} \forall E}{\frac{Gt}{\forall x Gx} \forall I} \wedge E_R$$

$$\frac{\forall x Fx \quad \forall x Gx}{(\forall x Fx) \wedge (\forall x Gx)} \wedge I$$

(d) $\forall xFx \wedge \forall xGx \vdash \forall x(Fx \wedge Gx)$

$$\frac{\frac{(\forall x Fx) \wedge (\forall x Gx)}{\forall x Fx} \wedge E_L}{\frac{Ft}{\forall x Fx} \forall E} \wedge E_L \quad \frac{\frac{(\forall x Fx) \wedge (\forall x Gx)}{\forall x Gx} \wedge E_R}{\frac{Gt}{\forall x Gx} \forall E} \wedge E_R$$

$$\frac{Ft \wedge Gt}{\forall x (Fx \wedge Gx)} \forall I$$

(e) $\forall x(P \rightarrow Fx) \vdash P \rightarrow \forall xFx$

$$\frac{\frac{\frac{\forall x(P \rightarrow Fx)}{P \rightarrow Ft} \forall E \quad \overline{P}^1}{\text{MP}}}{\frac{\frac{Ft}{\forall x Fx} \forall I}{P \rightarrow \forall x Fx} \rightarrow I, 1}$$

(f) $P \rightarrow \forall xFx \vdash \forall x(P \rightarrow Fx)$

$$\frac{\overline{P}^1 \quad \frac{P \rightarrow \forall x Fx}{\forall x Fx} \text{MP}}{\frac{\frac{Ft}{\forall x Fx} \forall E}{P \rightarrow Ft} \rightarrow I, 1} \forall I$$

(g) $\exists x(P \rightarrow Fx) \vdash P \rightarrow \exists xFx$

$$\frac{\frac{\frac{\overline{P \rightarrow Ft}^2 \quad \overline{P}^1}{\text{MP}}}{\frac{Ft}{\exists x Fx} \exists I} \rightarrow I, 2}{\frac{\exists x(P \rightarrow Fx) \quad (P \rightarrow Ft) \rightarrow \exists x Fx}{\frac{\exists x Fx}{P \rightarrow \exists x Fx} \rightarrow I, 1} \exists E}$$

(h) $\exists x \neg Fx \vdash \neg \forall x Fx$

$$\frac{\frac{\frac{\overline{\neg Ft}^1 \quad \frac{\overline{\forall x Fx}^2}{Ft} \forall E}{\text{MP}}}{\frac{\perp}{\neg Ft \rightarrow \perp} \rightarrow I, 1} \text{MP}}{\frac{\perp}{\neg \forall x Fx} \rightarrow I, 2}$$

(i) $\forall x \neg Fx \vdash \neg \exists x Fx$

$$\frac{\frac{\overline{\exists x Fx}^1 \quad \frac{\forall x \neg Fx}{Ft \rightarrow \perp} \forall E}{\text{MP}}}{\frac{\perp}{\neg \exists x Fx} \rightarrow I, 1}$$

(j) $\neg \exists x Fx \vdash \forall x \neg Fx$

$$\frac{\frac{\overline{Ft}^1}{\exists x Fx} \exists I \quad \neg \exists x Fx}{\text{MP}} \frac{\perp}{\neg Ft} \rightarrow I, 1 \forall I$$

(k) $\exists x Fx \rightarrow P \vdash \forall x(Fx \rightarrow P)$

Note that the binding conventions mean: $\exists x Fx \rightarrow P$ is $(\exists x Fx) \rightarrow P$. That is, $\exists$ binds tighter than $\rightarrow$ implication.

$$\frac{\frac{\overline{Ft}^1}{\exists x Fx} \exists I \quad \exists x Fx \rightarrow P}{\text{MP}} \frac{P}{Ft \rightarrow P} \rightarrow I, 1 \forall I$$

(l) $\exists x(Fx \rightarrow Gx) \vdash \forall xFx \rightarrow \exists xGx$

$$\frac{\frac{\frac{\overline{\forall x Fx}^1}{Ft} \forall E \quad \overline{Ft \rightarrow Gt}^2}{Gt} \exists I \quad \frac{\exists x (Fx \rightarrow Gx) \quad \frac{\overline{\exists x Gx}}{(Ft \rightarrow Gt) \rightarrow \exists x Gx} \rightarrow I, 2}{\frac{\overline{\exists x Gx}}{\forall x Fx \rightarrow \exists x Gx} \rightarrow I, 1} \exists E$$

(m) $\forall xFx \vdash \neg \exists x \neg Fx$

$$\frac{\frac{\overline{\neg Fx}^2 \quad \frac{\overline{\forall x Fx}}{Fx} \forall E}{\perp} \text{MP} \quad \frac{\overline{\exists x \neg Fx}^1 \quad \frac{\perp}{\neg Fx \rightarrow \perp} \rightarrow I, 2}{\perp} \text{MP} \quad \frac{\perp}{\neg \exists x \neg Fx} \rightarrow I, 1$$

(n) $\exists xFx \vdash \neg \forall x \neg Fx$

$$\frac{\frac{\overline{\exists x Fx} \quad \frac{\overline{\forall x \neg Fx}}{Fx \rightarrow \perp} \forall E}{\perp} \text{MP} \quad \frac{\perp}{\neg \forall x \neg Fx} \rightarrow I, 1$$

(o) $\forall x (Fx \rightarrow \neg Gx) \vdash \neg \exists x (Fx \wedge Gx)$

$$\frac{\frac{\overline{\forall x (Fx \rightarrow \neg Gx)}}{Fx \rightarrow \neg Gt} \forall E \quad \frac{\overline{Ft \wedge Gt}^2}{Ft} \wedge E_L \quad \frac{\overline{Ft \wedge Gt}^2}{Gt} \wedge E_R}{\frac{\overline{\neg Gt}}{Ft \wedge Gt \rightarrow \perp} \text{MP} \quad \frac{\perp}{Ft \wedge Gt \rightarrow \perp} \rightarrow I, 2} \exists E \quad \frac{\overline{\exists x (Fx \wedge Gx)}^1 \quad \frac{\perp}{\neg \exists x (Fx \wedge Gx)} \rightarrow I, 1$$

(p)

$$\vdash \exists x (Fx \vee Gx) \leftrightarrow \exists x Fx \vee \exists x Gx$$

To prove an $\leftrightarrow$ we prove each implication separately. They should be connected with an $\wedge I$ and introduction to complete the proof.

First we prove $\exists x (Fx \vee Gx) \vdash \exists x Fx \vee \exists x Gx$

$$\frac{\frac{\frac{\overline{Ft} \ 2}{\exists x Fx} \exists I \quad \frac{\overline{Gt} \ 3}{\exists x Gx} \exists I}{\frac{\overline{Ft} \vee \overline{Gt} \ 1 \quad \frac{\frac{\overline{Ft} \vee \overline{Gt}}{\exists x Fx \vee \exists x Gx} \vee I_L \rightarrow I, 2 \quad \frac{\overline{Ft} \vee \overline{Gt}}{\exists x Fx \vee \exists x Gx} \vee I_R \rightarrow I, 3}{\frac{\overline{Ft} \vee \overline{Gt} \rightarrow (\exists x Fx \vee \exists x Gx)}{\exists x Fx \vee \exists x Gx} \vee E} \rightarrow I, 1}{\frac{\exists x (Fx \vee Gx) \quad \frac{\overline{Ft} \vee \overline{Gt} \rightarrow (\exists x Fx \vee \exists x Gx)}{\exists x Fx \vee \exists x Gx} \rightarrow I, 1}{\exists x Fx \vee \exists x Gx} \exists I}$$

Next we prove $\exists x Fx \vee \exists x Gx \vdash \exists x (Fx \vee Gx)$

$$\frac{\frac{\frac{\overline{Ft} \ 2}{Ft \vee Gt} \vee I_L \quad \frac{\overline{Gt} \ 4}{Ft \vee Gt} \vee I_R}{\frac{\overline{Ft} \vee \overline{Gt}}{\exists x (Fx \vee Gx)} \exists I} \rightarrow I, 2 \quad \frac{\frac{\overline{Ft} \vee \overline{Gt}}{\exists x (Fx \vee Gx)} \exists I}{\frac{\overline{Ft} \vee \overline{Gt} \rightarrow \exists x (Fx \vee Gx)}{\exists x (Fx \vee Gx)} \exists E} \rightarrow I, 4}{\frac{\frac{\overline{Ft} \vee \overline{Gt} \rightarrow \exists x (Fx \vee Gx)}{\exists x (Fx \vee Gx)} \exists E \quad \frac{\overline{Ft} \vee \overline{Gt} \rightarrow \exists x (Fx \vee Gx)}{\exists x (Fx \vee Gx)} \exists E}{\frac{\exists x Fx \vee \exists x Gx \quad \frac{\overline{Ft} \vee \overline{Gt} \rightarrow \exists x (Fx \vee Gx)}{\exists x (Fx \vee Gx)} \rightarrow I, 1 \quad \frac{\overline{Ft} \vee \overline{Gt} \rightarrow \exists x (Fx \vee Gx)}{\exists x (Fx \vee Gx)} \rightarrow I, 3}{\exists x (Fx \vee Gx)} \vee E}$$

2. Prove the following in the predicate calculus. Ex falso or classical modes of reasoning (RAA, LEM, DNE) will be helpful for proving these theorems. Each of these are challenging!

(a) $\forall x (Fx \vee Gx), \forall x \neg Gx \vdash \forall x Fx$

This is the universally quantified version of the “disjunctive syllogism” from the Lab on intuitionistic logic. Once the quantifiers have been eliminated, then we can simply call on this theorem from propositional logic - that is what the rule DS is denoting.

$$\frac{\frac{\forall x (Fx \vee Gx)}{Ft \vee Gt} \forall E \quad \frac{\forall x \neg Gx}{\neg Gt} \forall E}{\frac{Ft}{\forall x Fx} \forall I} \text{DS}$$

(b) $\neg \forall x Fx \vdash \exists x \neg Fx$

We will use RAA to prove this. Which means we can prove the following sequent instead, before using RAA to prove the original sequent:

$$\frac{\frac{\frac{\neg \forall x Fx, \neg \exists x \neg Fx \vdash \perp}{\neg \forall x Fx} \text{RAA}, 2}{\exists x \neg Fx} \text{MP}}{\frac{\frac{\frac{\frac{\frac{\frac{\perp}{\neg \neg Fa} \rightarrow I, 1}{\neg Fa} \text{DNE}}{Fa} \forall I}{\forall x Fx} \forall I}{\neg \forall x Fx} \text{MP}}{\perp} \text{RAA}, 2} \text{MP}$$

(c) $\neg \forall x \neg Fx \vdash \exists x Fx$

$$\frac{\frac{\frac{\frac{\frac{\frac{\perp}{\neg Ft} \rightarrow I, 1}{\neg Ft} \forall I}{\forall x \neg Fx} \forall I}{\neg \forall x \neg Fx} \text{MP}}{\perp} \text{RAA}, 2}{\exists x Fx} \text{MP}$$

(d) $\neg \exists x \neg Fx \vdash \forall x Fx$

$$\frac{\frac{\frac{\frac{\perp}{\neg Ft} \text{RAA}, 1}{Ft} \forall I}{\exists x \neg Fx} \text{MP}}{\forall x Fx} \text{MP}$$

The final $\forall I$ is valid as t has been discharged in the previous step.

Note: The exercises above show that in the presence of classical modes of reasoning it is sufficient to introduce only one of the quantifiers, as the other can be deduced from it. Sequents above show that we could define $\forall = \neg \exists \neg$ and $\exists = \neg \forall \neg$ in classical logic. Following the BHK, however, one needs to define both independently.